Negative density dependence promotes persistence of a globally rare yet locally abundant plant species *(Oenothera coloradensis)*

Supporting Information

Species Information

Oenothera coloradensis seeds are contained within small, indehiscent capsules that contain 2-5 seeds each (Burgess, Hild, & Shaw, 2005). A single adult individual can produce >500 capsules. This species does not reproduce vegetatively, although seeds typically germinate near the base of the parent plant, which often results in dense patches of vegetative individuals (Heidel, Tuthill, & Wallace, 2021). Previous work established that *O. coloradensis* population growth rate is particularly impacted by recruitment of seedlings (Floyd & Ranker, 1998). Recruitment increases when non-*O. coloradensis* community biomass is removed, indicating that surrounding grasses and forbs outcompete or shade-out seedlings (Munk, Hild, & Whitson, 2002).

O. coloradensis commonly co-occurs with *Agrostis stolonifera*, *Pascopyrum smithii*, *Poa pratensis*, *Glycyrrhiza lepidota*, *Iris missouriensis*, *Cirsium flodmanii*, and *Grindelia squarrosa* (U.S. Fish & Wildlife Service, 2019; Munk et al., 2002). Encroachment of woody shrubs such as *Salix exigua* has been correlated with declining numbers in some populations (Heidel et al., 2021).

The Wyoming Natural Diversity Database (WYNDD) began a base-wide census of reproductive individuals in the FEWAFB population in 1986, and has repeated this census annually since 1988 (Heidel et al., 2021). The first estimate of species size after its full

geographic range was identified occurred in 1998, when it was approximated that the entire species consisted of 47,300 to 50,300 reproductive individuals (Fertig, 2000).

Continuous Vital Rate Models

We created "global" models for each continuous vital rate (Table 1 in the main text). We modeled survival probability ($s(z)$) as a function of log-transformed leaf $size_t$ using generalized linear models with binomial error distributions. Flowering individuals were excluded from the data used to fit survival models, since *O. coloradensis* is a monocarpic perennial that nearly always dies after flowering. We modeled probability of flowering ($Pb(z)$) using generalized linear models with binomial error distributions, and was predicted by log-transformed leaf $size_t$ plus log-transformed leaf $size_t$ squared. This squared term was initially in other vital rate models also, but its inclusion was not supported by AIC model selection. We modeled seed production ($b(z)$) as a function of $size_t$ using negative binomial models because the count data was over-dispersed. We only used data from flowering plants to fit these models, using the "glm.nb" function from the "MASS" R package (Venables & Ripley, 2002). We described growth, or the distribution of plant size in year $t+1$ ($G(z', z)$) as a series of Normal distributions with mean = μ_s and standard deviation = σ_s. μ_s was modeled as a function of $size_t$ using a linear model with Gaussian error. σ_s was the residual standard error of this linear model. We described the distribution of recruit size in $year_{t+1}$ ($c_o(z')$) as a Normal distribution with the mean μ_r, and the standard deviation σ_r. μ_r and σ_r were the mean and standard deviation of observed longest leaf size in year $t+1$.

Discrete Vital Rate Parameter Estimation

Previously-published data from a greenhouse experiment using *O. coloradensis* seed capsules collected from the FEWAFB populations determined that viable seeds had an average germination rate of 20.3% after cold-stratification, and did not decline in germination rate over five years (Burgess et al., 2005). This study also found that 58.5% of seeds produced were viable. We conducted an additional seed study to determine if overwintering in natural conditions lead to a lower germination rate than was identified in the previous greenhouse study. We buried 60 field-collected seed capsules in mesh bags at 6 locations near our demographic study plots at FEWAFB, and then recovered the seed bags after one winter. An average of 10% of seed capsules were not recoverable, likely because they were non-viable and withered away or were eaten. We planted the recovered capsules in standard greenhouse conditions, and found a mean germination rate of 6.8%. This germination rate was much lower than that identified by Burgess et al., however our seed study had a much smaller sample size, reducing the reliability of our result (Burgess et al., 2005). However, it is still likely that true germination rates are much lower than those identified in greenhouse conditions, so we reduced the germination rate identified by Burgess et al. by 20%.

Table S2: Continuous vital rate functions used in each IPM. In the functions below, $size_t$ indicates plant longest leaf length in the previous year, N_t indicates number of individuals in a plot in the previous year, T_G indicates mean temperature in the previous year's growing season, and T_W indicates the mean temperature in winter of the previous year.

IPM	Vital Rate	Equation
A & B	Survival	$logit(s(z)) = -0.21 + 0.37(ln(size_t))$
	Flowering	$\text{logit}(Pb(z)) = -34.06 + 27.67(ln(size_t)) - 5.78(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.57 + 0.18(ln(size_t))$; $\sigma_s = 0.51$
	Seed prod.	$exp(b(z)) = 3.35 + 1.19(ln(size_t))$

	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.21$; $\sigma_r = 0.77$
C	Survival	$logit(s(z)) = -0.73 + 0.45(ln(size_t))$
	Flowering	$logit(Pb(z)) = -40.31 + 33.40(ln(size_t)) - 7.09(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.64 + 0.11(ln(size_t))$; $\sigma_s = 0.42$
	Seed prod.	$exp(b(z)) = 3.73 + 1.01(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.19$; $\sigma_r = 0.76$
D	Survival	$logit(s(z)) = 1.01 + -0.01(ln(size_t))$
	Flowering	$logit(Pb(z)) = -44.52 + 36.73(ln(size_t)) - 7.69(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.84 + 0.26(ln(size_t))$; $\sigma_s = 0.52$
	Seed prod.	$exp(b(z)) = 3.80 + 1.11(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.22$; $\sigma_r = 0.78$
E	Survival	$logit(s(z)) = 0.21 + 0.18(ln(size_t))$
	Flowering	$logit(Pb(z)) = -18.73 + 15.84(ln(size_t)) - 3.63(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.74 + 0.24(ln(size_t))$; $\sigma_s = 0.49$
	Seed prod.	$exp(b(z)) = 3.83 + 0.92(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.19$; $\sigma_r = 0.80$
F	Survival	$logit(s(z)) = 0.54 + 0.24(ln(size_t))$
	Flowering	$logit(Pb(z)) = -27.49 + 20.83(ln(size_t)) - 4.04(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.62 + 0.16(ln(size_t))$; $\sigma_s = 0.44$
	Seed prod.	$exp(b(z)) = 1.72 + 1.51(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.15$; $\sigma_r = 0.80$
G	Survival	$logit(s(z)) = -0.19 + 0.30(ln(size_t))$
	Flowering	$logit(Pb(z)) = -32.72 + 23.50(ln(size_t)) - 4.23(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.42 + 0.21(ln(size_t))$; $\sigma_s = 0.42$
	Seed prod.	$exp(b(z)) = 0.83 + 2.06(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.21$; $\sigma_r = 0.77$
H	Survival	$logit(s(z)) = -0.32 + 0.62(ln(size_t))$
	Flowering	$logit(Pb(z)) = -31.04 + 23.96(ln(size_t)) - 4.71(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.50 + 0.05(ln(size_t))$; $\sigma_s = 0.43$
	Seed prod.	$exp(b(z)) = 2.53 + 1.59(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.23$; $\sigma_r = 0.76$
I	Survival	$logit(s(z)) = -0.33 + 0.46(ln(size_t)) - 0.0003(N_t)$
	Flowering	$logit(Pb(z)) = -41.36 + 33.73(ln(size_t)) - 7.21(ln(size_t)^2) + 0.0008(N_t)$

	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 2.19 + 0.14(ln(size_t)) - 0.0005(N_t)$; $\sigma_s = 0.42$
	Seed prod.	$exp(b(z)) = 3.73 + 1.01(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.19$; $\sigma_r = 0.76$
J	Survival	$logit(s(z)) = 17.59 + 0.05(ln(size_t)) - 0.03(N_t)$
	Flowering	$logit(Pb(z)) = -46.27 + 37.38(ln(size_t)) - 7.83(ln(size_t)^2) + 0.002(N_t)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 3.70 + 0.28(ln(size_t)) - 0.004(N_t)$; $\sigma_s = 0.51$
	Seed prod.	$exp(b(z)) = 3.80 + 1.11(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.22$; $\sigma_r = 0.78$
K	Survival	$logit(s(z)) = -13.85 + 0.20(ln(size_t)) + 0.04(N_t)$
	Flowering	$logit(Pb(z)) = -21.21 + 15.50(ln(size_t)) - 3.55(ln(size_t)^2) + 0.009(N_t)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = -0.49 + 0.24(ln(size_t)) + 0.006(N_t)$; $\sigma_s = 0.48$
	Seed prod.	$exp(b(z)) = 3.83 + 0.92(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.19$; $\sigma_r = 0.80$
L	Survival	$logit(s(z)) = 33.06 + 0.24(ln(size_t)) - 0.44(N_t)$
	Flowering	$logit(Pb(z)) = -26.23 + 21.0(ln(size_t)) - 4.01(ln(size_t)^2) - 0.02(N_t)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 15.89 + 0.17(ln(size_t)) - 0.19(N_t)$; $\sigma_s = 0.43$
	Seed prod.	$exp(b(z)) = 1.72 + 1.51(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.15$; $\sigma_r = 0.80$
M	Survival	$logit(s(z)) = 5.39 + 0.37(ln(size_t)) - 0.02(N_t)$
	Flowering	$logit(Pb(z)) = -32.79 + 24.02(ln(size_t)) - 4.34(ln(size_t)^2) - 0.002(N_t)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 2.26 + 0.23(ln(size_t)) - 0.003(N_t)$; $\sigma_s = 0.39$
	Seed prod.	$exp(b(z)) = 0.83 + 2.06(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.21$; $\sigma_r = 0.77$
N	Survival	$logit(s(z)) = 4.30 + 0.87(ln(size_t)) - 0.004(N_t)$
	Flowering	$logit(Pb(z)) = -28.50 + 25.52(ln(size_t)) - 4.93(ln(size_t)^2) - 0.004(N_t)$

	Growth	$G(z',z) = N(\mu_s, \sigma_s)$; $\mu_s = 2.76 + 0.16(ln(size_t)) - 0.001(N_t)$; $\sigma_s = 0.36$
	Seed prod.	$exp(b(z)) = 2.53 + 1.59(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.23$; $\sigma_r = 0.77$
S	Survival	$logit(s(z)) = 1.11 + 0.49(ln(size_t)) + 0.005(N_t) - 0.28(T_G)$
	Flowering	$logit(Pb(z)) = -47.12 + 33.70(ln(size_t)) - 7.21(ln(size_t)^2) + 0.0004(N_t) + 0.42(T_G) - 0.23(T_W)$
	Growth	$G(z',z) = N(\mu_s, \sigma_s)$; $\mu_s = 4.12 + 0.14(ln(size_t)) + 0.0009(N_t) - 0.20(T_G)$; $\sigma_s = 0.41$
	Seed prod.	$exp(b(z)) = 3.52 + 0.90(ln(size_t)) + 0.0008(N_t) + 0.01(T_G) + 0.10(T_W)$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = -0.38 + 0.00005(N_t) + 0.04(T_G) + 0.03(T_W)$; $\sigma_r = 0.76$
T	Survival	$logit(s(z)) = -27.59 + 0.21(ln(size_t)) - 0.0006(N_t) + 1.91(T_G)$
	Flowering	$logit(Pb(z)) = -70.28 + 35.37(ln(size_t)) - 7.45(ln(size_t)^2) - 0.003(N_t) + 1.83(T_G) - 0.95(T_W)$
	Growth	$G(z',z) = N(\mu_s, \sigma_s)$; $\mu_s = -2.88 + 0.27(ln(size_t)) + 0.0002(N_t) + 0.32(T_G)$; $\sigma_s = 0.49$
	Seed prod.	$exp(b(z)) = -12.0 + 0.80(ln(size_t)) - 0.002(N_t) + 1.07(T_G) - 0.94(T_W)$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = -0.65 - 0.0007(N_t) + 0.06(T_G) - 0.16(T_W)$; $\sigma_r = 0.77$
U	Survival	$logit(s(z)) = -14.02 + 0.39(ln(size_t)) + 0.006(N_t) + 0.87(T_G)$
	Flowering	$logit(Pb(z)) = -22.83 + 16.0(ln(size_t)) - 3.66(ln(size_t)^2) + 0.002(N_t) + 0.19(T_G) - 1.01(T_W)$
	Growth	$G(z',z) = N(\mu_s, \sigma_s)$; $\mu_s = 0.40 + 0.25(ln(size_t)) - 0.001(N_t) + 0.10(T_G)$; $\sigma_s = 0.46$
	Seed prod.	$exp(b(z)) = -14.33 + 0.59(ln(size_t)) + 0.001(N_t) + 1.18(T_G) - 1.17(T_W)$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = -0.56 - 0.00002(N_t) + 0.05(T_G) + 0.09(T_W)$; $\sigma_r = 0.81$
V	Survival	$logit(s(z)) = -6.78 + 0.24(ln(size_t)) - 0.08(N_t) + 0.69(T_G)$
	Flowering	$logit(Pb(z)) = -36.50 + 18.80(ln(size_t)) - 3.57(ln(size_t)^2) - 0.02(N_t) + 0.79(T_G) + 0.09(T_W)$

	Growth	$G(z',z) = N(\mu_s, \sigma_s)$; $\mu_s = -1.83 + 0.17(ln(size_t)) + 0.0003(N_t) + 0.23(T_G)$; $\sigma_s = 0.42$
	Seed prod.	$exp(b(z)) = 8.44 + 1.32(ln(size_t)) + 0.003(N_t) - 0.43(T_G) - 0.18(T_W)$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = -2.01 + 0.006(N_t) + 0.13(T_G) - 0.01(T_W)$; $\sigma_r = 0.80$
W	Survival	$logit(s(z)) = -32.30 + 0.38(ln(size_t)) + 0.0008(N_t) + 2.19(T_G)$
	Flowering	$logit(Pb(z)) = -311.80 + 18.72(ln(size_t)) - 3.56(ln(size_t)^2) - 0.002(N_t) + 17.93(T_G) - 20.46(T_W)$
	Growth	$G(z',z) = N(\mu_s, \sigma_s)$; $\mu_s = -2.73 + 0.22(ln(size_t)) - 0.0006(N_t) + 0.29(T_G)$; $\sigma_s = 0.39$
	Seed prod.	$exp(b(z)) = -7.07 + 1.47(ln(size_t)) + 0.003(N_t) + 0.63(T_G)$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 1.75 + 0.006(N_t) - 0.10(T_G) + 0.17(T_W)$; $\sigma_r = 0.77$
X	Survival	$logit(s(z)) = -34.59 + 0.88(ln(size_t)) + 0.001(N_t) + 2.26(T_G)$
	Flowering	$logit(Pb(z)) = -43.64 + 28.21(ln(size_t)) - 5.35(ln(size_t)^2) - 0.001(N_t) + 0.57(T_G) + 1.33(T_W)$
	Growth	$G(z',z) = N(\mu_s, \sigma_s)$; $\mu_s = -7.65 + 0.16(ln(size_t)) - 0.0000004(N_t) + 0.61(T_G)$; $\sigma_s = 0.36$
	Seed prod.	$exp(b(z)) = -28.65 + 0.62(ln(size_t)) + 0.0005(N_t) + 2.15(T_G) - 1.64(T_W)$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.48 - 0.00004(N_t) - 0.01(T_G) + 0.005(T_W)$; $\sigma_r = 0.77$
AA	Survival	$logit(s(z)) = -0.26 + 0.54(ln(size_t))$
	Flowering	$logit(Pb(z)) = -28.96 + 21.71(ln(size_t)) - 4.14(ln(size_t)^2)$
	Growth	$G(z',z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.48 + 0.10(ln(size_t))$; $\sigma_s = 0.44$
	Seed prod.	$exp(b(z)) = 3.12 + 1.24(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.23$; $\sigma_r = 0.77$
BB	Survival	$logit(s(z)) = -0.18 + 0.29(ln(size_t))$
	Flowering	$logit(Pb(z)) = -32.70 + 26.93(ln(size_t)) - 5.73(ln(size_t)^2)$
	Growth	$G(z',z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.72 + 0.18(ln(size_t))$; $\sigma_s = 0.51$
	Seed prod.	$exp(b(z)) = 3.45 + 1.17(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.19$; $\sigma_r = 0.77$
CC	Survival	$logit(s(z)) = -1.23 + 0.76(ln(size_t))$
	Flowering	$logit(Pb(z)) = -45.81 + 38.00(ln(size_t)) - 8.03(ln(size_t)^2)$

	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.57 + 0.12(ln(size_t))$; $\sigma_s = 0.44$
	Seed prod.	$exp(b(z)) = 3.21 + 1.22(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.13$; $\sigma_r = 0.77$
DD	Survival	$logit(s(z)) = 2.43 - 0.23(ln(size_t))$
	Flowering	$logit(Pb(z)) = -86.83 + 71.52(ln(size_t)) - 14.65(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.93 + 0.25(ln(size_t))$; $\sigma_s = 0.56$
	Seed prod.	$exp(b(z)) = 6.01 + 0.34(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.30$; $\sigma_r = 0.69$
EE	Survival	$logit(s(z)) = 0.65 + 0.15(ln(size_t))$
	Flowering	$logit(Pb(z)) = -24.35 + 17.91(ln(size_t)) - 3.49(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.85 + 0.20(ln(size_t))$; $\sigma_s = 0.52$
	Seed prod.	$exp(b(z)) = 2.51 + 1.54(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.09$; $\sigma_r = 0.86$
FF	Survival	$logit(s(z)) = 1.50 - 0.32(ln(size_t))$
	Flowering	$logit(Pb(z)) = -70.62 + 54.30(ln(size_t)) - 10.46(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.65 + 0.21(ln(size_t))$; $\sigma_s = 0.46$
	Seed prod.	$exp(b(z)) = 1.91 + 1.39(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.10$; $\sigma_r = 0.92$
GG	Survival	$logit(s(z)) = 0.99 + 0.12(ln(size_t))$
	Flowering	$logit(Pb(z)) = -50.32 + 36.50(ln(size_t)) - 6.54(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.48 + 0.28(ln(size_t))$; $\sigma_s = 0.38$
	Seed prod.	$exp(b(z)) = 3.06 + 1.22(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.18$; $\sigma_r = 0.81$
HH	Survival	$logit(s(z)) = 0.74 + 0.33(ln(size_t))$
	Flowering	$logit(Pb(z)) = -50.91 + 40.17(ln(size_t)) - 7.81(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.62 + 0.18(ln(size_t))$; $\sigma_s = 0.39$
	Seed prod.	$exp(b(z)) = 4.20 + 0.97(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.24$; $\sigma_r = 0.76$
II	Survival	$logit(s(z)) = -0.26 + 0.06(ln(size_t))$
	Flowering	$logit(Pb(z)) = -41.11 + 36.347(ln(size_t)) - 8.31(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.67 + 0.16(ln(size_t))$; $\sigma_s = 0.38$
	Seed prod.	$exp(b(z)) = 4.92 + 0.41(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.22$; $\sigma_r = 0.71$
JJ	Survival	$logit(s(z)) = 0.07 + 0.16(ln(size_t))$

	Flowering	$logit(Pb(z)) = -24.83 + 19.82(ln(size_t)) - 4.20(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.63 + 0.32(ln(size_t))$; $\sigma_s = 0.44$
	Seed prod.	$exp(b(z)) = 4.63 + 0.66(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.09$; $\sigma_r = 0.82$
KK	Survival	$logit(s(z)) = -0.26 + 0.25(ln(size_t))$
	Flowering	$logit(Pb(z)) = -27.90 + 27.66(ln(size_t)) - 7.00(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.56 + 0.30(ln(size_t))$; $\sigma_s = 0.41$
	Seed prod.	$exp(b(z)) = 5.72 - 0.09(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.80$; $\sigma_r = 0.79$
LL	Survival	$logit(s(z)) = -0.22 + 0.77(ln(size_t))$
	Flowering	$logit(Pb(z)) = -147.87 + 121.01(ln(size_t)) - 24.82(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.61 + 0.10(ln(size_t))$; $\sigma_s = 0.39$
	Seed prod.	$exp(b(z)) = 1.98 + 1.57(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.16$; $\sigma_r = 0.68$
MM	Survival	$logit(s(z)) = -1.13 + 0.60(ln(size_t))$
	Flowering	$logit(Pb(z)) = -45.56 + 39.31(ln(size_t)) - 8.64(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.34 + 0.14(ln(size_t))$; $\sigma_s = 0.40$
	Seed prod.	$exp(b(z)) = 0.71 + 2.00(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.32$; $\sigma_r = 0.64$
NN	Survival	$logit(s(z)) = -1.74 + 1.39(ln(size_t))$
	Flowering	$logit(Pb(z)) = -27.89 + 20.50(ln(size_t)) - 3.99(ln(size_t)^2)$
	Growth	$G(z', z) = N(\mu_s, \sigma_s)$; $\mu_s = 1.17 + 0.13(ln(size_t))$; $\sigma_s = 0.33$
	Seed prod.	$exp(b(z)) = 5.49 + 0.14(ln(size_t))$
	Recruit size	$c_o(z') = N(\mu_r, \sigma_r)$; $\mu_r = 0.24$; $\sigma_r = 0.78$

Table S1: Permanent Plot Locations and subpopulation-level sample sizes for each year and individual type (seedling vs. non-seedling). GPS coordinates listed in decimal degrees, map datum and spheroid: WGS 84.

					Sample Size					
					2018		2019		2020	
Site	Subpopulation	Plot Name	N Coord.	W Coord.	non-seedling	seedling	non-seedling	seedling	non-seedling	seedling
FEWAFB	Unnamed Creek	U3	41.13642	-104.87209	740	525	528	417	406	530
	Unnamed Creek	U4	41.13634	-104.87183						
	Unnamed Creek	U6	41.13647	-104.87132						
	Diamond Creek	D7	41.14340	-104.88380	235	209	347	149	275	81
	Diamond Creek	D10	41.14441	-104.88303						
	Diamond Creek	D11	41.14431	-104.88094						
	Crow Creek	C4	41.15540	-104.87497	203	127	214	98	150	160
	Crow Creek	C5	41.15477	-104.87474						
	Crow Creek	C8	41.15534	-104.87487						
Soapstone	Pasture HQ5	S1	40.99297	-105.00925	283	772	714	813	641	423
	Pasture HQ5	S2	40.99318	-105.00935						
	Pasture HQ5	S3	40.99342	-105.00937						
	Pasture HQ3	S4	40.98623	-105.01691	102	138	158	173	117	104
	Pasture HQ3	S5	40.98639	-105.01671						
	Pasture HQ3	S6	40.98650	-105.01656						
	Meadow	S7	40.98753	-105.02148	44	31	47	28	48	12
	Meadow	S8	40.98747	-105.02179						
	Meadow	S9	40.98724	-105.02145						

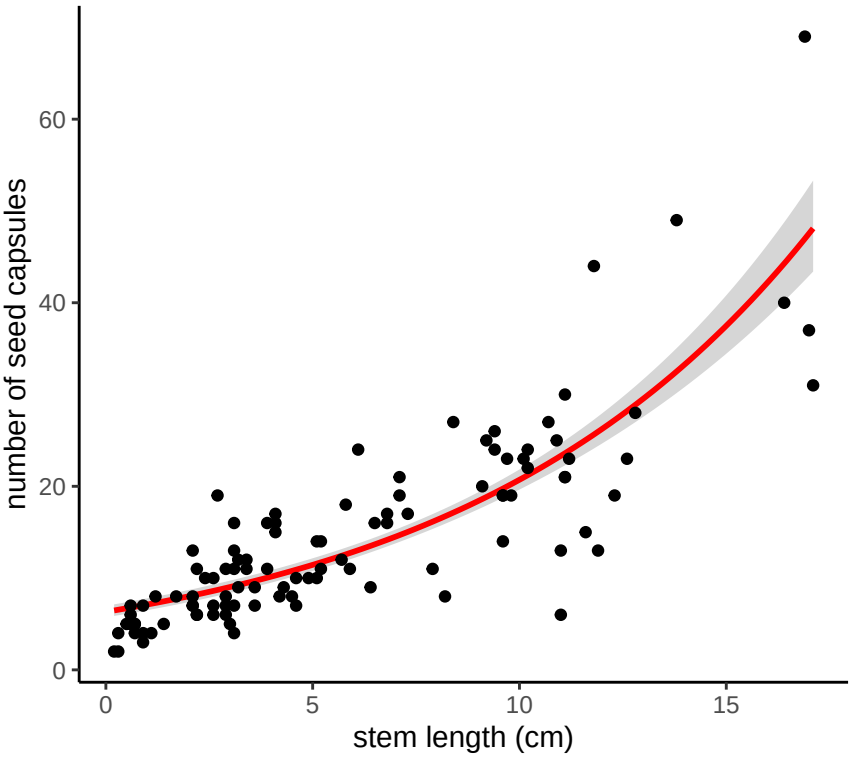


Figure S1: As the stem length of an *Oenothera coloradensis* flowering individual increases, the number of capsules it produces increases as well. The red line shows the fit from a Poisson generalized linear model, and the grey ribbon shows the 95% confidence interval around the fitted relationship. Model equation: Number of capsules $= e^{(1.843+0.119 \times S))}$, where S is stem length in cm (pseudo R-squared $= 0.42$, $P = < 0.01$, Residual deviance $= 186.98$, df $= 104$).

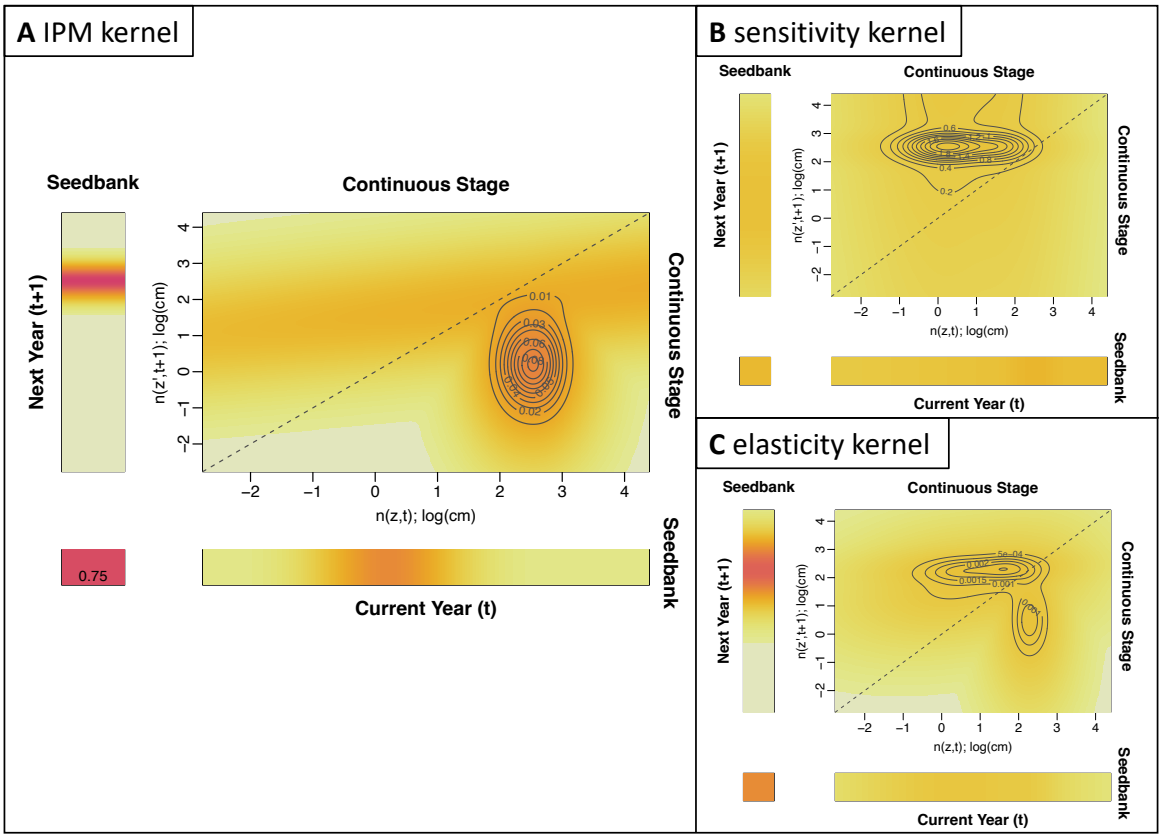


Figure S2: Visualizations of the *O. coloradensis* IPM kernels. (**A**) The IPM kernel for *O. coloradensis*. This kernel shows a density-independent IPM constructed using all data from all transitions (IPM "B"). (**B**) Sensitivity of the IPM kernel. (**C**) Elasticity of the IPM kernel. In all panels, color indicates probability, with darker colors corresponding to higher probability, and lighter colors corresponding to lower probability. The dashed line shows a 1:1 line. Contour lines in all panels indicate the locations of highest values in each plot, and are labeled with the corresponding values (Panel **A** = probability, panel **B** = sensitivity, panel **C** = elasticity).

References

Burgess, L. M., Hild, A. L., & Shaw, N. L. (2005). Capsule treatments to enhance seedling emergence of *Gaura neomexicana* ssp. *coloradensis*. *Restoration Ecology*, *13*(1), 8–14. https://doi.org/ 10.1111/j.1526-100X.2005.00002.x

Fertig, W. (2000). *Status review of the Colorado butterfly plant (*Gaura neomexicana *ssp.* coloradensis*)* (Tech. Rep.). Laramie, WY: Prepared for U.S. Fish and Wildlife Service and F. E. Warren Air Force Base by the Wyoming Natural Diversity Database (University of Wyoming).

Floyd, S. K., & Ranker, T. A. (1998). Analysis of a transition matrix model for *Gaura neomexicana* ssp. *coloradensis* (Onagraceae) reveals spatial and temporal demographic variability. *International Journal of Plant Sciences*, *159*(5), 853–863.

Heidel, B., Tuthill, D., & Wallace, Z. (2021). *33-Year population trends of Colorado butterfly plant (*Oenothera coloradensis*; Onagraceae), a Short-Lived Riparian Species on F. E. Warren Air Force Base, Laramie County, Wyoming* (Tech. Rep.). Laramie, WY: Prepared for U.S. Fish and Wildlife Service and F. E. Warren Air Force Base by the Wyoming Natural Diversity Database (University of Wyoming).

Krakos, K. (2013). Personal communication with Bonnie Heidel.

Munk, L. M., Hild, A. L., & Whitson, T. D. (2002). Rosette recruitment of a rare endemic forb (*Gaura neomexicana* subsp. *coloradensis*) with canopy removal of associated species. *Restoration Ecology*, *10*(1), 122–128. https://doi.org/ 10.1046/j.1526-100X.2002.10113.x

U.S. Fish & Wildlife Service. (2019). *Endangered and Threatened Wildlife and Plants; Removing* Oenothera coloradensis *(Colorado butterfly plant) from the federal list of endangered and threatened plants.* 50 C.F.R. pt. 17. (https://www.federalregister.gov/documents/2019/11/05/2019-24124/endangered-and-threatened-wildlife-and-plants-removing-oenothera-coloradensis-colorado-butterfly)

Venables, W. N., & Ripley, B. D. (2002). *Modern Applied Statistics with S* (Fourth ed.). New York: Springer. Retrieved from `https://www.stats.ox.ac.uk/pub/MASS4/`